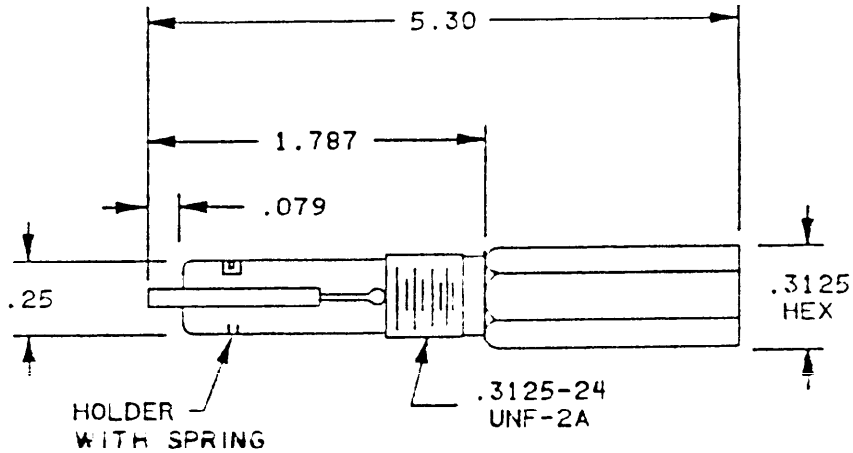
MILITARY SPECIFICATION SHEET

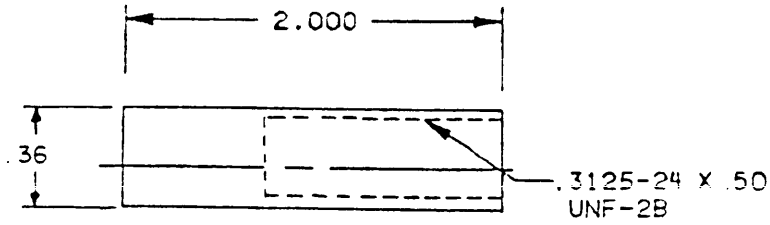
TOOLS, FIBER OPTIC, SCRIBE, DIAMOND OR SAPPHIRE, TYPE I

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring fiber optic tools described herein shall consist of this specification and the latest issue of MIL-T-83523.

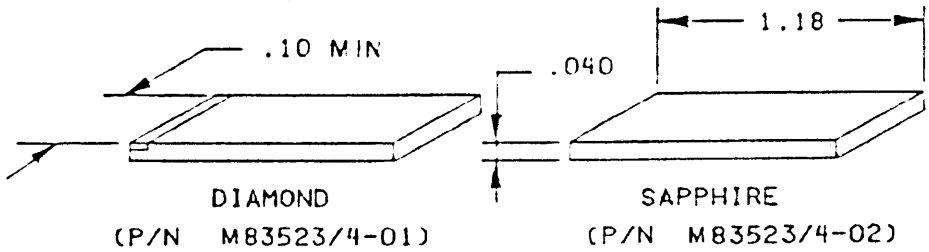


DIAMOND AND SAPPHIRE
SCRIBE ASSEMBLY



Inches	mm
.040	1.02
.079	2.01
.10	2.5
.25	6.4
.3125	7.938
.36	9.144
.50	12.7
1.18	30.0
1.787	45.39
2.000	50.80
5.30	134.6

PROTECTIVE CAP /4



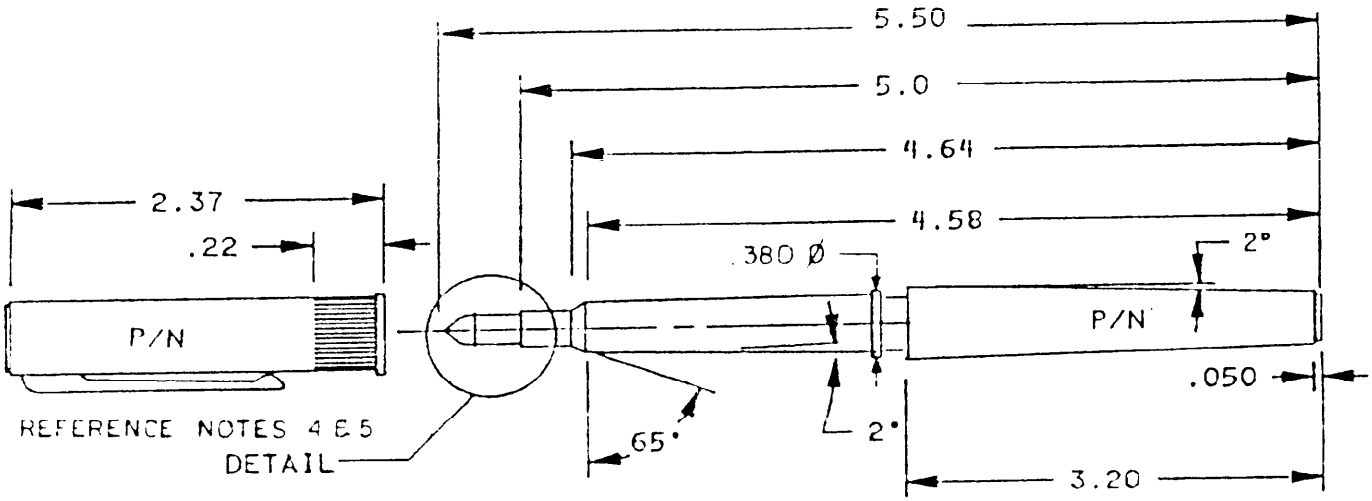
(P/N M83523/4-01) (P/N M83523/4-02)

Blades

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.01 (0.3 mm) for two place decimals. Angles ±1°.
4. Wall thickness of protective cap .047 minimum.

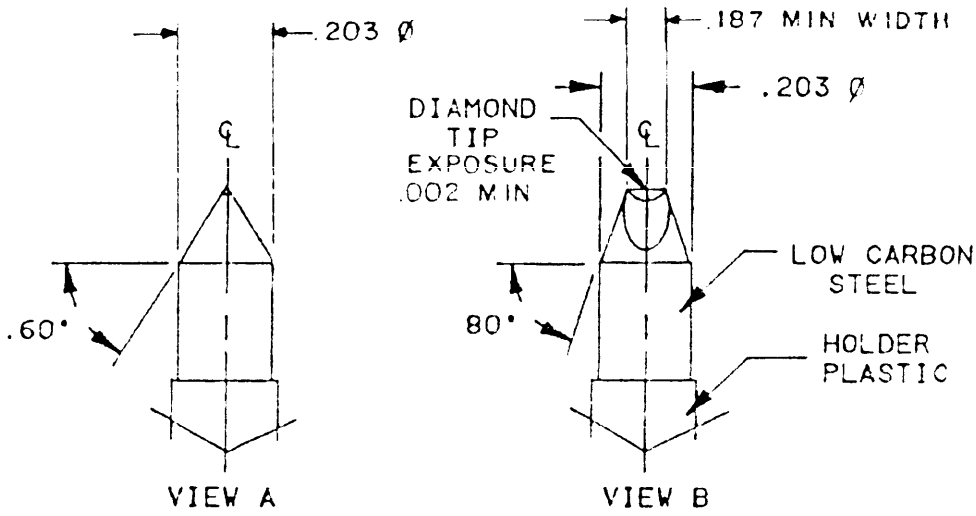
FIGURE 1. Fiber optic diamond and sapphire scribe with replacement blades.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:36Z
Check the source to verify that this is the current version before use.



CAP - PRESS FIT

Inches	mm
.002	0.051
.050	1.27
.187	4.75
.203	5.16
.220	5.59
.380	9.65
2.375	60.33
3.20	81.28
4.58	116.33
4.64	117.86
5.0	127.00
5.50	139.70



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.01 (0.3 mm) for two place decimals. Angles ±1°.
4. Wall thickness of protective cap .065 minimum.
5. Cap should snap fit over .380 dimension.

FIGURE 2. Fiber optic diamond scribe (P/N M83523/4-03).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:36Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2.

Weight: 4 ounces maximum.

Material:

Part numbers M83523/4-01 and M83523/4-02: Blades are mounted into a holder in the tool and are interchangeable and adjustable. Tools come with protective snap-on or screw type caps. The holders shall be molybdenum in accordance with AMS 2465B-83 or equivalent.

Part number M83523/4-03: Scribe tool tip shall be made from a natural industrial diamond containing no chip inclusions or carbon spots on or near the cutting edge as verified at 60X magnification. The holder shall be made of molded plastic with a removable cap used for protection when the tool is not in use.

Engagement of all threaded parts must be at least two full threads.

Part number: See table I.

Application: These tools shall be used to scribe optical fibers as specified in DOD-F-49291.

TABLE I. Part numbers.

Type	Tool
Diamond Sapphire	M83523/4-01 M83523/4-02
Diamond	M83523/4-03

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, GL, MI
Navy - AS, OS, SH
Air Force - 11, 12, 13, 14, 70, 71, 80, 82, 84, 90
DLA - ES

User activities:
Army - AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6010-0016)

Source: https://assist.dla.mil -- Downloaded: 202610-23T08:36Z
Check the source to verify that this is the currenversion before use.